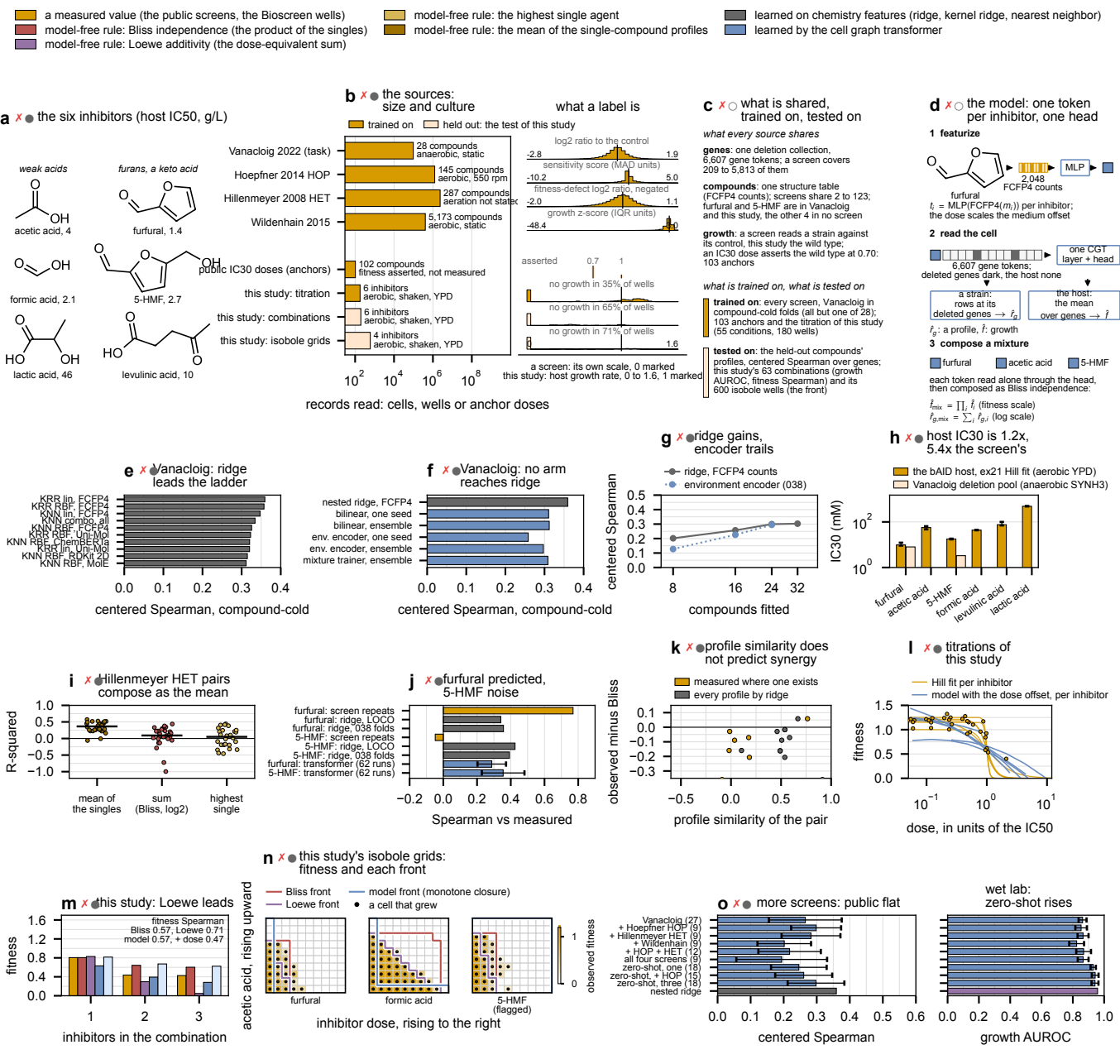

a measured value (the public screens, the Bioscreen wells)
model-free rule: Bliss independence (the product of the singles)
model-free rule: Loewe additivity (the dose-equivalent sum)
model-free rule: the highest single agent
model-free rule: the mean of the single-compound profiles
learned on chemistry features (ridge, kernel ridge, nearest neighbor)
learned by the cell graph transformer
a ✗ ● the six inhibitors (host IC50, g/L)
weak acids
furans, a keto acid
acetic acid, 4
furfural, 1.4
formic acid, 2.1
5-HMF, 2.7
lactic acid, 46
levulinic acid, 10
b ✗ ● the sources: size and culture
what a label is
trained on
held out: the test of this study
Vanacloig 2022 (task)
28 compounds anaerobic, static
Hoepfner 2014 HOP
145 compounds aerobic, 550 rpm
Hillenmeyer 2008 HET
287 compounds aeration not static
Wildenhain 2015
5,173 compounds aerobic, static
public IC30 doses (anchors)
102 compounds fitness asserted, not measured
this study: titration
6 inhibitors aerobic, shaken, YPD
this study: combinations
6 inhibitors aerobic, shaken, YPD
this study: isobole grids
4 inhibitors aerobic, shaken, YPD
records read: cells, wells or anchor doses
log2 ratio to the control
sensitivity score (MAD units)
fitness-defect log2 ratio, negated
growth z-score (IQR units)
asserted
no growth in 35% of wells
no growth in 65% of wells
no growth in 71% of wells
a screen: its own scale, 0 marked
this study: host growth rate, 0 to 1.6, 1 marked
c ✗ ○ what is shared, trained on, tested on
what every source shares
genes: one deletion collection, 6,607 gene tokens; a screen covers 209 to 5,813 of them
compounds: one structure table (FCFP4 counts); screens share 2 to 123; furfural and 5-HMF are in Vanacloig and this study, the other 4 in no screen
growth: a screen reads a strain against its control, this study the wild type; an IC30 dose asserts the wild type at 0.70: 103 anchors
what is trained on, what is tested on
trained on: every screen, Vanacloig in compound-cold folds (all but one of 28); 103 anchors and the titration of this study (55 conditions, 180 wells)
tested on: the held-out compounds' profiles, centered Spearman over genes; this study's 63 combinations (growth AUROC; fitness Spearman) and its 600 isobole wells (the front)
d ✗ ○ the model: one token per inhibitor, one head
1 featurize
furfural
MLP
2,048 FCFP4 counts
t_i = MLP(FCFP4(m_i)) per inhibitor; the dose scales the medium offset
2 read the cell
6,607 gene tokens; deleted genes dark, the host none
one CGT layer + head
a strain: rows at its deleted genes → $\hat{r}_g$
the host: the mean over genes → $\hat{t}$
$\hat{r}_g$: a profile, $\hat{t}$: growth
3 compose a mixture
furfural
acetic acid
5-HMF
each token read alone through the head, then composed as Bliss independence:
$\hat{f}_{mix} = \prod_i \hat{f}_i$ (fitness scale)
$\hat{r}_{g,mix} = \sum_i \hat{r}_{g,i}$ (log scale)
e ✗ Vanacloig: ridge leads the ladder
KRR lin, FCFP4
KRR RBF, FCFP4
KNN lin, FCFP4
KNN comps, all
KNN RBF, FCFP4
KRR RBF, Uni-Mol
KNN RBF, ChemBERTa
KRR lin, Uni-Mol
KNN RBF, RDKit 2D
KNN RBF, MolE
centered Spearman, compound-cold
f ✗ Vanacloig: no arm reaches ridge
nested ridge, FCFP4
bilinear, one seed
bilinear, ensemble
env. encoder, one seed
env. encoder, ensemble
mixture trainer, ensemble
centered Spearman, compound-cold
g ✗ ridge gains, encoder trails
ridge, FCFP4 counts
environment encoder (038)
centered Spearman
compounds fitted
h ✗ ● host IC30 is 1.2x, 5.4x the screen's
the bAID host, ex21 Hill fit (aerobic YPD)
Vanacloig deletion pool (anaerobic SYNH3)
IC30 (mM)
furfural
acetic acid
5-HMF
formic acid
levulinic acid
lactic acid
i ✗ Hillenmeyer HET pairs compose as the mean
R-squared
mean of the singles
sum (Bliss, log2)
highest single
j ✗ furfural predicted, 5-HMF noise
furfural: screen repeats
furfural: ridge, LOCO
furfural: ridge, 038 folds
5-HMF: screen repeats
5-HMF: ridge, LOCO
5-HMF: ridge, 038 folds
furfural: transformer (62 runs)
5-HMF: transformer (62 runs)
Spearman vs measured
k ✗ ● profile similarity does not predict synergy
measured where one exists
every profile by ridge
observed minus Bliss
profile similarity of the pair
l ✗ ● titrations of this study
Hill fit per inhibitor
model with the dose offset, per inhibitor
fitness
dose, in units of the IC50
m ✗ this study: Loewe leads
fitness Spearman
Bliss 0.57, Loewe 0.71
model 0.57, + dose 0.47
fitness
inhibitors in the combination
n ✗ ● this study's isobole grids: fitness and each front
Bliss front
Loewe front
model front (monotone closure)
a cell that grew
acetic acid, rising upward
furfural
formic acid
5-HMF (flagged)
observed fitness
inhibitor dose, rising to the right
o ✗ ● more screens: public flat
Vanacloig (27)
+ Hoepfner HOP (9)
+ Hillenmeyer HET (9)
+ Widenhain (9)
+ HOP + HET (12)
all four screens (9)
zero-shot, one (18)
zero-shot, + HOP (15)
zero-shot, three (18)
nested ridge
centered Spearman
wet lab: zero-shot rises
growth AUROC